



مدينة زويل للعلوم والتكنولوجيا
Zewail City of Science and Technology

Thermofluids

(NANENG 218)

Project report

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Spring 2026

Rapid thermal processing (RTP)



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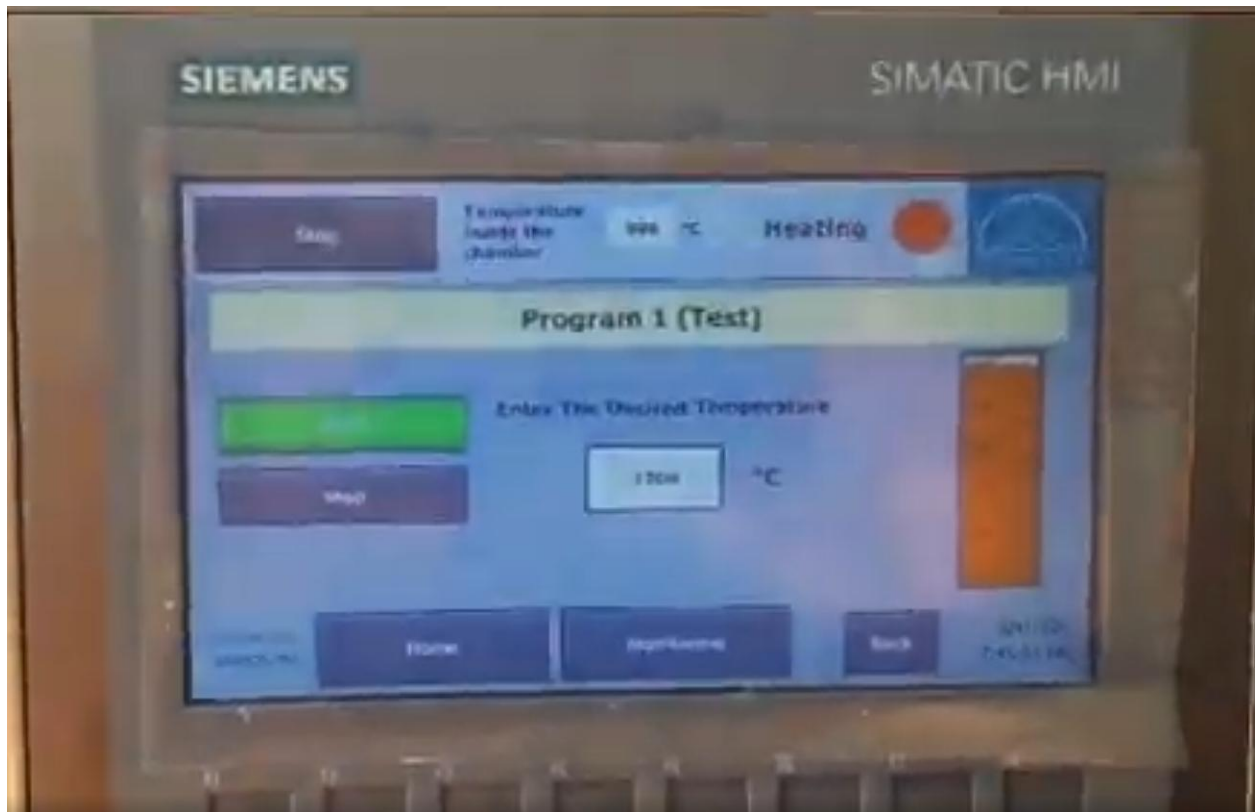
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Introduction:

Rapid Thermal Processing (RTP) is a semiconductor manufacturing process that involves heating silicon wafers to high temperatures sometimes exceeding 1,000°C at a short time (few seconds). This process is used for many applications in semiconductor manufacturing, including dopant activation, thermal oxidation, and chemical vapor deposition. This is done by using either high-intensity lamps or lasers, the wafers are heated fast during RTP, and then cooled slowly to avoid dislocations and thermal shock-related wafer breaking. Measuring and controlling the wafer temperature during RTP using pyrometry is a major challenge.

Importance:

Rapid Thermal Processing (RTP) plays a very important role in semiconductor manufacturing because it provides rapid and precise thermal treatments required for producing high-quality semiconductor devices. RTP systems offer very fast heating and cooling cycles, which improve manufacturing speed and efficiency while reducing processing time. They also provide accurate temperature control, which is essential for maintaining device uniformity and achieving reliable semiconductor performance. Another major advantage of RTP is the reduction of the thermal budget, since wafers spend less time at high temperatures, helping prevent unwanted dopant diffusion and preserving device structures. In addition, RTP is highly versatile and can be used in many semiconductor processes such as annealing, oxidation, chemical vapor deposition, and dopant activation. The controlled thermal treatments provided by RTP improve the electrical characteristics of semiconductor devices, enhance dopant activation, and improve gate oxide quality. Furthermore, RTP systems can be scaled for large-scale industrial production and integrated easily into modern semiconductor fabrication lines, making RTP an essential technology in advanced semiconductor manufacturing.



(A device that can control RTP process at Zewail city labs)

Concepts, mechanism, and design:

Concepts:

- Rapid Heating Mechanisms

The defining characteristic of RTP is the ability to heat a wafer at rates far exceeding those achievable in conventional furnaces. Heating in RTP systems is accomplished primarily through radiant energy transfer from high-intensity light sources. The silicon wafer, being a semiconductor, absorbs photons with energies above its bandgap (~ 1.12 eV at room temperature), converting them directly into heat. The dominant heating mechanisms include:

- **Photon Absorption:** Radiative absorption from tungsten-halogen lamps or arc lamps emitting primarily in the near-infrared (NIR) spectrum (0.7–3 μm), where silicon absorbs strongly.
- **Temperature-Dependent Absorption:** The emissivity and absorptivity of the wafer are temperature-dependent, complicating accurate temperature control. At lower temperatures, silicon is more transparent to longer wavelengths; at higher temperatures, free-carrier absorption increases.
- **Thermal Conduction:** Heat conduction within the wafer redistributes absorbed energy. The thermal conductivity of silicon decreases with temperature, affecting temperature uniformity across the wafer.

Mechanisms:

- Rapid Cooling Mechanisms

Unlike heating, cooling in RTP relies on radiation from the wafer surface and conduction to the surrounding chamber gas. Since RTP chambers are designed for rapid heating, active cooling is sometimes employed to accelerate the cooling rate after the thermal cycle. Key cooling mechanisms include:

- **Radiative Emission:** The wafer radiates heat according to the Stefan-Boltzmann law. At temperatures above 500°C, radiative cooling is significant.
- **Gas Convection:** The chamber gas carries heat away from the wafer surface via natural and forced convection (in real life application).

- Thermal Cycle Profiles

An RTP thermal cycle is defined by four principal parameters:

Parameter	Description	Typical Range
Ramp-Up Rate	Rate of temperature increase from standby to process temperature	10–300°C/s
Process Temperature	Peak temperature at which the desired reaction occurs	700–1200°C
Soak Time	Duration at process temperature	0–60 seconds (spike: <1 s)
Ramp-Down Rate	Rate of temperature decrease after soak	30–150°C/s (passive)

- Temperature Measurement

Accurate, non-contact temperature measurement is essential in RTP. The primary technique is optical pyrometry, which infers temperature from the thermal radiation emitted by the wafer surface. However, this requires knowledge of the wafer's spectral emissivity, which varies with:

- Wafer composition (bulk silicon, thin films, metal layers).
- Surface roughness and texture.
- Temperature itself.

- Lamp Array Configurations

The spatial arrangement of lamps relative to the wafer is a central design decision in RTP systems. Common configurations include:

- **Dual-Side Irradiation:** Both top and bottom lamp arrays irradiate the wafer simultaneously, providing symmetric heating and reducing through-wafer temperature gradients. This is the most common configuration for production RTP tools.

- Chamber Design

The RTP chamber serves multiple functions: it contains the process ambient, provides a mounting surface for lamps and wafer supports, and acts as a reflector to enhance heating efficiency. Key design features include:

- **Reflective Chamber Walls:** Quartz chambers transmit the NIR radiation from tungsten-halogen lamps efficiently and can withstand repeated thermal cycling.

- Temperature Control Systems

Closed-loop temperature control is implemented using pyrometer feedback to adjust lamp power. Advanced control strategies include:

- **PID Control:** PID (Proportional-Integral-Derivative) controllers are widely used for their simplicity.

Key challenges:

-Temperature Uniformity

Achieving uniform temperature across a wafer (during rapid heating and cooling) is the foremost technical challenge in RTP. Temperature non-uniformity leads to spatially varying dopant activation, film properties, and silicide formation, all of which degrade device performance and yield.

- Process Scalability

RTP processes must deliver consistent results across millions of wafers in high-volume manufacturing. Challenges to scalability include:

- **Lamp Aging:** Lamp aging changes the spectral output and intensity distribution over time, shifting the temperature calibration.
- **Chamber Aging:** Repeated thermal cycling causes degradation of quartz components, reflector coatings, and edge rings, altering chamber optical properties.

-Integration with Existing Fabrication Processes

RTP must be integrated seamlessly into complex, multi-step fabrication flows. Challenges include:

- **Material Compatibility:** The RTP step must be compatible with films and structures deposited in preceding steps. Some materials (e.g., photoresist residues, low- κ dielectrics) cannot tolerate high temperatures.

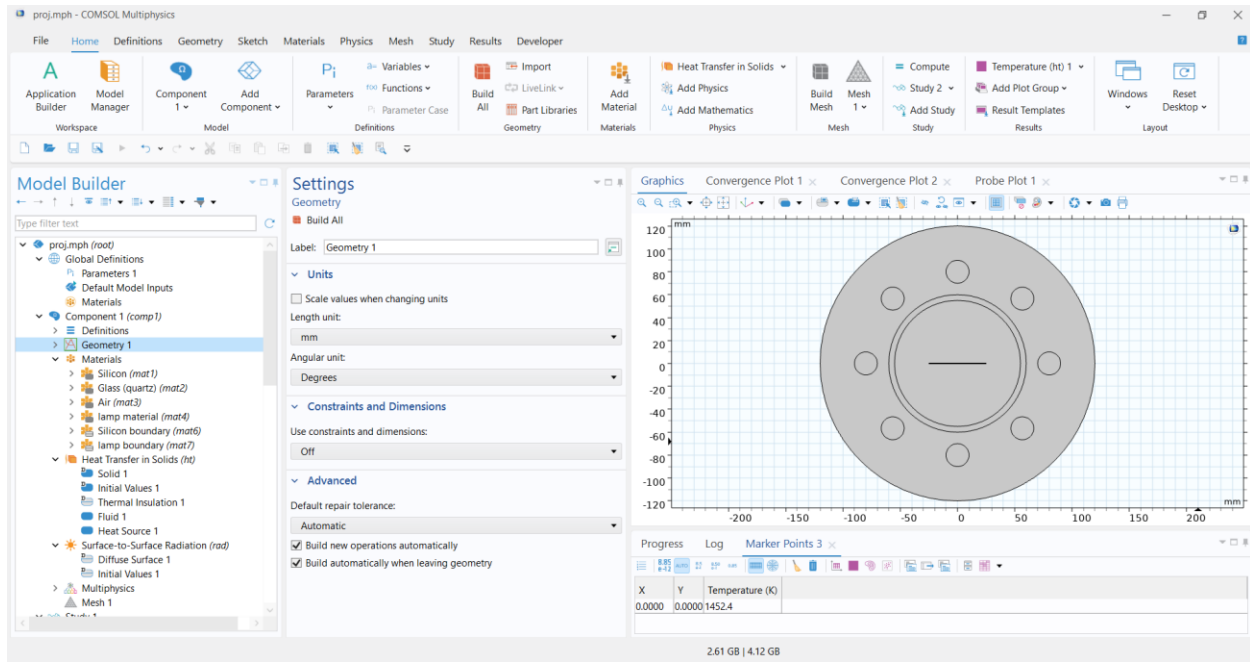
Components and reason of choosing them:

RTP System Components

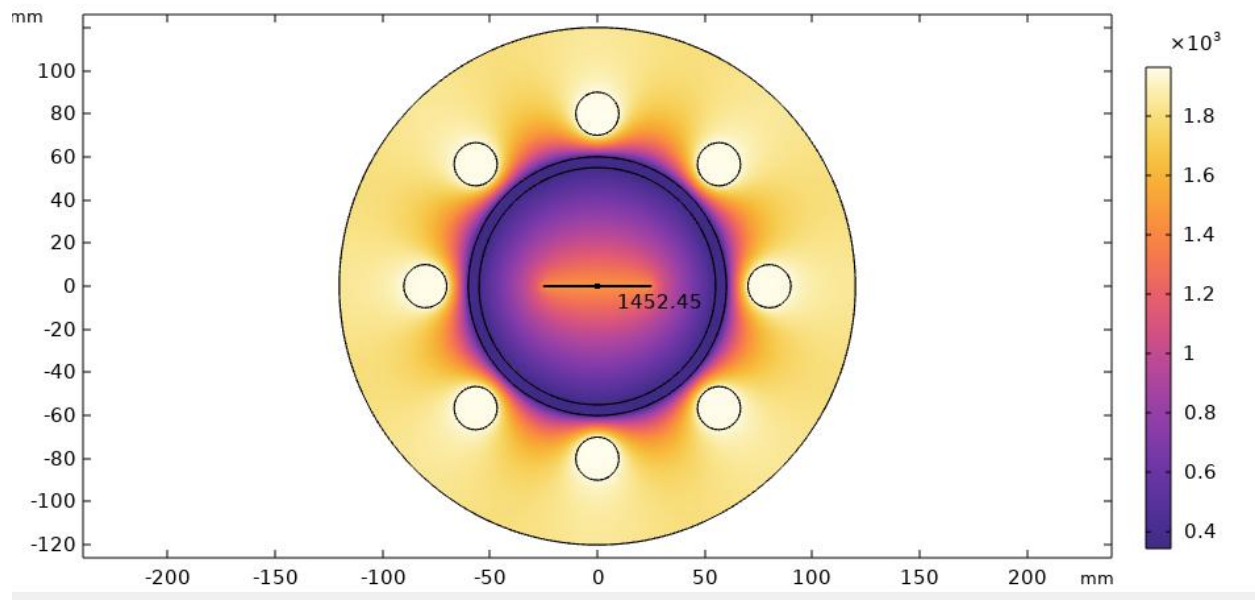
- Heat Source: Tungsten is commonly used in RTP chambers for its high melting point, and it provides high temperatures up to 1000°C rapidly including IR radiation.
- Light Tube: Made of quartz (SiO₂) because it is transparent for IR light and can handle high temperatures. It doesn't also contribute to any contamination.
- Wafer Support: To hold the wafer during the process, it is required to handle high temperature as well hence it is made of graphite. Graphite helps in providing uniform heating.
- Cooling Management: Cools the wafer after delivering high temperatures to avoid any thermal shock. It is usually made of ceramics to avoid any electric conduction.

Comsol results and analysis:

Here is the geometry and the materials used in a COMSOL design to study the effect of the RTP:

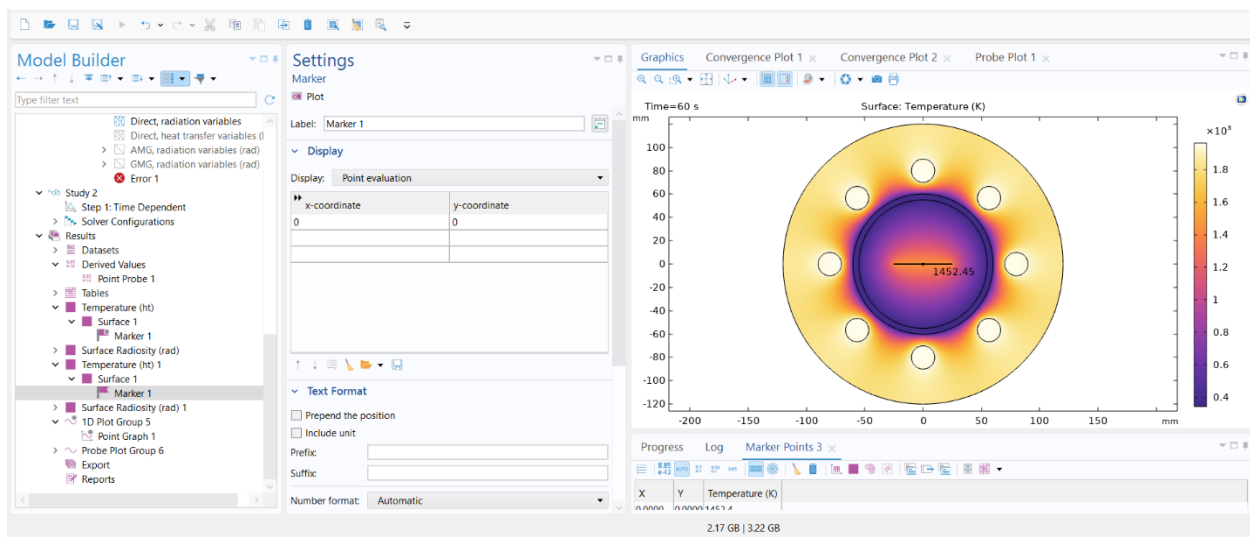


This is the steady state temperature distribution:

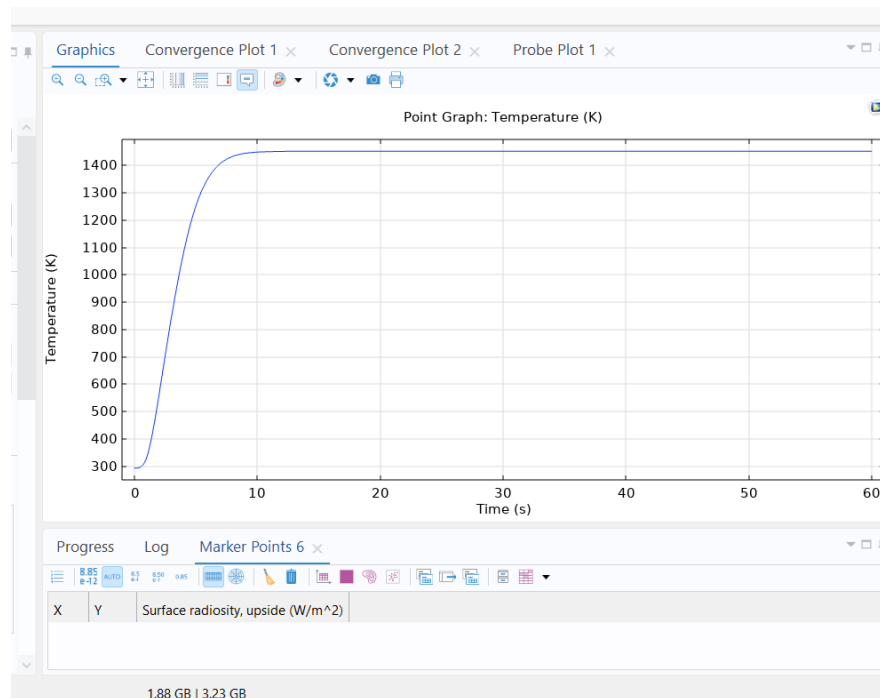


The simulation plot demonstrates the steady-state thermal profile of a Rapid Thermal Processing (RTP) chamber governed by solid heat conduction and non-linear surface-to-surface radiation. The eight radially symmetric infrared (IR) lamps serve as the primary heat source, generating peak temperatures near 1,900 K and radiating energy across the chamber to heat the central silicon wafer. While the geometric center of the wafer successfully reaches an industrially optimal processing temperature of 1,452.45 K (approximately 1,179°C) suitable for standard semiconductor fabrication steps like rapid thermal annealing, the profile highlights a critical engineering challenge regarding thermal uniformity. Due to direct proximity to the heating ring, the outer edges of the wafer experience a much higher radiation view factor and localized heat flux than the core. The center must rely on internal thermal conduction to distribute this arriving energy, creating a steep thermal gradient across the substrate that would cause structural wafer warping or non-uniform film growth in a real manufacturing environment.

Temperature vs time (time varying study):



This is the plot of temperature vs time at the midpoint(0,0):



1. The Initial Thermal Inertia (0 to 1 Second)

At the absolute start ($t = 0$ s), there is a very brief, subtle lag before the temperature spikes. This represents the thermal mass and inertia of the silicon wafer. Even though 40,000 W of radiative power is instantly turned on, the atoms in the solid lattice require a fraction of a second to absorb the initial photon flux and convert it into kinetic vibrational energy.

2. The Rapid Radiative Ramp-Up (1 to 8 Seconds)

Following the initial lag, the curve exhibits a massive, nearly vertical slope ($\Delta T / \Delta t$). This is the signature advantage of an RTP system. Because the silicon wafer is thin (low thermal mass) and silicon has high absorptivity for short-wavelength infrared radiation, it captures the intense radiant energy from the lamp ring with extreme efficiency. The temperature skyrockets at a rate exceeding 150 K per second, traveling through the non-linear radiation regime without destabilizing the solver.

3. Thermodynamic Equilibrium Plateau (10 to 60 Seconds)

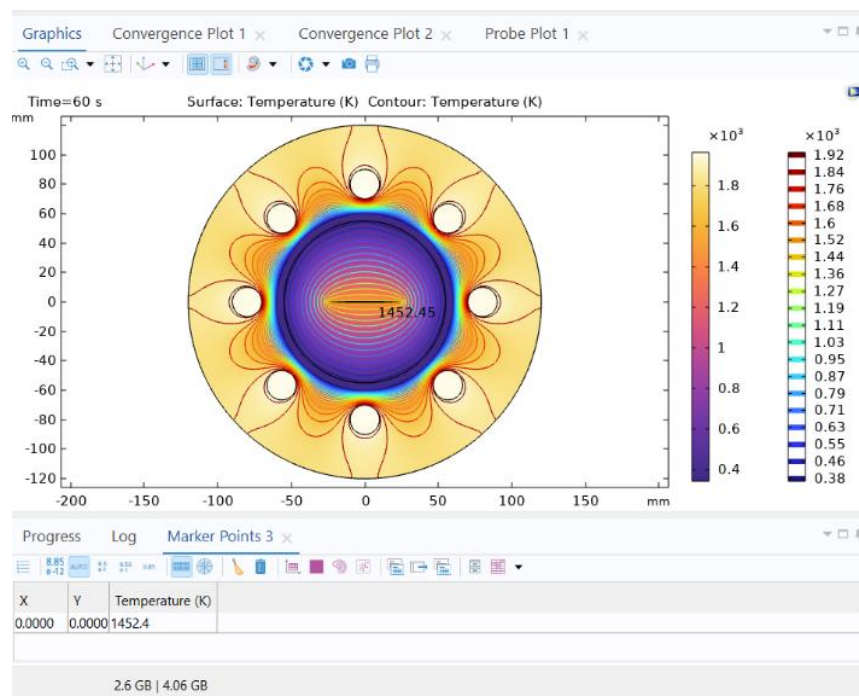
By approximately 10 seconds, the sharp temperature rise rounds off perfectly and enters a flat plateau. This stabilization represents the onset of steady-state thermal equilibrium. At this exact temperature (1,452.45 K), a perfect balance is achieved inside the chamber:

Energy Absorbed from Lamps = Energy Lost via Radiation + Internal Conduction

The wafer cannot get any hotter because the energy it sheds back out into the cooler chamber walls via radiation (T^4 dependencies) and internal conduction matches the energy coming in.

It proves that this model achieves its processing target temperature almost instantly, minimizing the overall "thermal budget" and preventing dopants from diffusing out of control in the silicon.

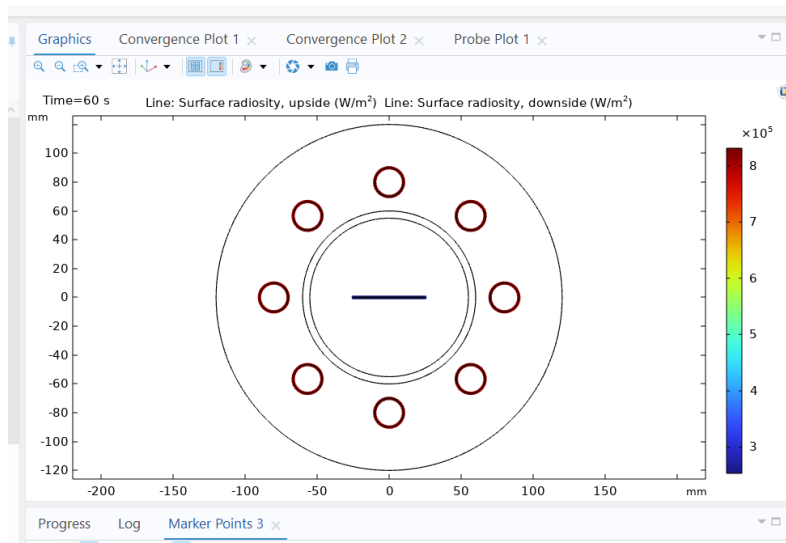
Isotherms:



The inclusion of an isothermal contour plot at $t = 60$ s provides a direct spatial map of the thermal gradients ($\text{grad } T$) within the RTP chamber. The high density of concentric contour lines surrounding the eight infrared lamps indicates severe localized temperature gradients at the primary heat sources. As the radiant energy

propagates past the quartz envelope, the expanding profile of the isotherms illustrates the directional path of the radiative heat flux. Along the central horizontal axis, the contour lines tightly intersect the silicon wafer, visually demonstrating the severe center-to-edge thermal non-uniformity. This profile confirms that while the geometric center has successfully stabilized at an industrially viable processing temperature of 1,452.45, the perimeter is subjected to a much higher direct heat flux, relying on internal lattice conduction to drive heat toward the cooler core.

Radiosity:



- The Intense Crimson Rings (The Lamps): The eight heating lamps show the highest radiosity on the scale, reaching up to 8×10^5 (W/m²). This confirms that they are the primary source of power in the system, furiously emitting massive amounts of short-wavelength infrared photons into the chamber.
- The Ghostly/Transparent Mediums (Air & Quartz): We can notice how the open air gaps and the quartz tube completely disappear or show zero radiosity values. This is perfect physical proof of transparency. Because the gas and quartz do not absorb or reflect the infrared light waves, the radiation passes straight through them without registering a radiosity value on their surfaces.
- The Dark Wafer Line in the Center: The silicon wafer displays a lower radiosity value relative to the blazing lamps. This happens because silicon has a high absorptivity—it swallows the incoming photon energy to heat itself up to 1,452.45

K rather than bouncing it back out into the chamber. The radiosity we do see on the wafer line is primarily its own high-temperature thermal emission.

Conclusion:

The simulation successfully models an RTP chamber where a silicon wafer reaches a stable processing temperature of 1,452.45 K (1,179 degC) within 10 seconds via infrared lamp radiation. However, surface radiosity and temperature contours reveal a steep center-to-edge gradient; the edges receive intense direct radiation while the center relies on slower internal conduction. In production, this severe thermal non-uniformity would cause catastrophic wafer warping, demonstrating the mechanical necessity for industrial wafer rotation or multi-zone lamp control.

References:

COMSOL Multiphysics®. (2022). *Rapid Thermal Annealing* (Application ID: 504). Heat Transfer Module, Application Gallery. COMSOL AB, Stockholm, Sweden. Retrieved from <https://www.comsol.com/model/rapid-thermal-annealing-504>

Roozeboom, F. (Ed.). (1996). *Advances in Rapid Thermal and Integrated Processing*. NATO ASI Series, Vol. 318. Kluwer Academic Publishers / Springer, Dordrecht. <link.springer.com/book/10.1007/978-94-015-8711-2>

Roozeboom, F. (1993). *Rapid Thermal Processing: Status, Problems and Options After the First 25 Years*. MRS Online Proceedings Library, Vol. 303, pp. 149. Cambridge Core. <doi.org/10.1557/PROC-303-149>

Dilhac, J.-M. (1996). *Temperature and Process Control in Rapid Thermal Processing*. In: *Advances in Rapid Thermal and Integrated Processing* (NATO ASI Series, Vol. 318). Springer, Dordrecht. doi.org/10.1007/978-94-015-8711-2_5

Norman, S. A., Schaper, C. D., & Boyd, S. P. (1991). *Improvement of Temperature Uniformity in Rapid Thermal Processing Systems Using Multivariable Control*. MRS Symposium Proceedings, Vol. 224, pp. 177–183. [Stanford University \(PDF\)](#)

Roozeboom, F. (1993). *Temperature Control and System Design Aspects in Rapid Thermal Processing*. MRS Online Proceedings Library, Vol. 224, pp. 9. Cambridge Core. <doi.org/10.1557/PROC-224-9>

Fiory, A. T. (2002). *Recent Developments in Rapid Thermal Processing*. *Journal of Electronic Materials*, 31(10), 981–987. Springer Nature. <doi.org/10.1007/s11664-002-0031-9>